ANALYSING BLUESKY DISCUSSION OF THE 2026 NEPAL FLOODS

Engagement, discussion themes and term networks

DOUNGSETH SAMANG
ARJAN SANDHU
OMAR HUSSAIN

WESTERN SYDNEY UNIVERSITY

RESEARCH AIM

This project investigates how the 2026 Nepal floods were discussed on Bluesky. The investigation examines whether posts containing external links received different engagement, identifies major discussion themes using clustering, and analyses relationships between important terms using network analysis.
Do posts containing links receive different engagement from posts without links?
What major themes appear in the discussion?
Which terms are most central within the discussion network?

DATASET

- Public Bluesky posts discussing the 2026 Nepal floods were collected using the Bluesky API through the atrrr package in R.
- Search queries: Nepal flood, Nepal floods, Nepal flooding, Nepal glacier
- Collection period: From 26 August 2026
- Initial records: 1,584
- Preprocessing: Duplicate removal, text cleaning, stop-word removal and stemming

METHOD

Hypothesis Testing:	Welch two-sample t-test comparing mean engagement between posts with links and posts without links.
Clustering:	TF-IDF weighting and unit-length normalisation were applied before K-means clustering. The elbow method was used to select (k = 7).
Network Analysis:	A term co-occurrence network was constructed using the top 500 TF-IDF terms. Degree and eigenvector centrality were used to identify influential terms.

HYPOTHESIS TEST

Do posts containing links receive different engagement?

Without links	With links
5.57 mean engagement	**5.00 mean engagement**

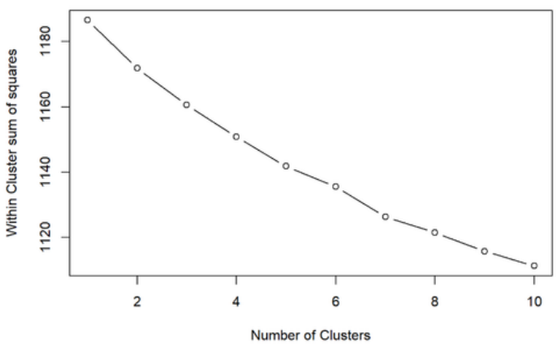
Welch two-sample t-test: p = 0.66

No statistically significant difference in mean engagement was found between posts with and without links.

RESULT

CLUSTERING

number of clusters vs SSW

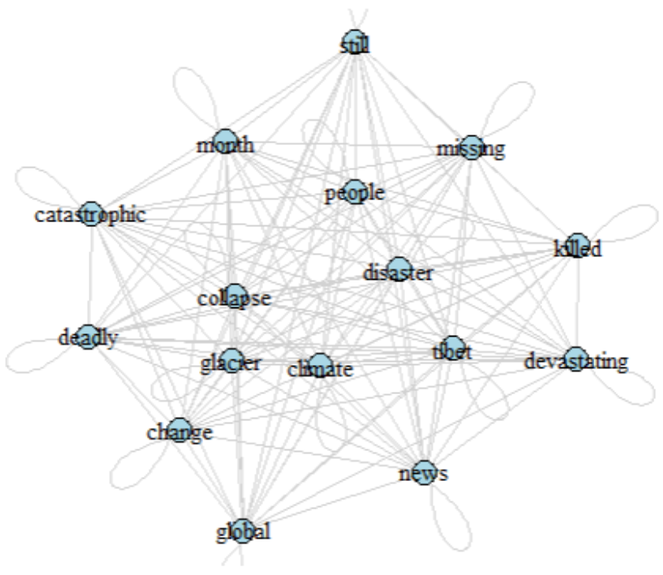


Seven clusters were selected using the elbow method.

Cluster <int>	Number.of.posts <dbl>	Top.Terms <chr>	Possible.Theme <chr>
1	80	landslid, india, kill, least	Landslides and casualties in India
2	75	warm, glacier, human, health	Warming, glaciers and human health
3	133	climat, justic, disast, fund	Climate justice and disaster funding
4	812	miss, disast, peopl, devast	General disaster impact and missing people
5	32	first, studi, tibet, crisi	Research and Tibet crisis discussion
6	63	chang, glacier, collaps, like	Climate change and glacier collapse
7	10	account, share, author, concern	Information sharing and account discussion

Cluster 4 was substantially larger than the other clusters and represented broad discussion of the disaster, including missing people, affected communities and devastation. Smaller clusters represented more specific environmental, humanitarian and climate-related themes.

NETWORK ANALYSIS



- The network analysis identified climate as the most central term. Other influential terms included disaster, glacier, change and collapse.
- Degree centrality
  Climate: 402
  Disaster: 357
- Eigenvector centrality
  Climate: 0.68
  Glacier: 0.28

These results suggest that environmental and climate-related concepts were strongly connected within discussion of the Nepal floods.

KEY FINDINGS AND LIMITATION

Key Findings: The presence of external links was not associated with a statistically significant difference in engagement (p = 0.66). K-means clustering identified seven discussion themes, with the largest cluster representing broad disaster impacts. Smaller clusters focused on landslides, casualties, glaciers, climate change, climate justice and disaster funding. Network analysis also identified climate-related terms as highly central, supporting the clustering results.

Limitations: The dataset contains only posts returned by selected Bluesky search terms and therefore may not represent all discussion of the Nepal floods or wider public opinion. Text preprocessing and stemming may also remove some linguistic detail.

Github LINK: https://github.com/samangdoungseth-oss/SWA_group_assignment